

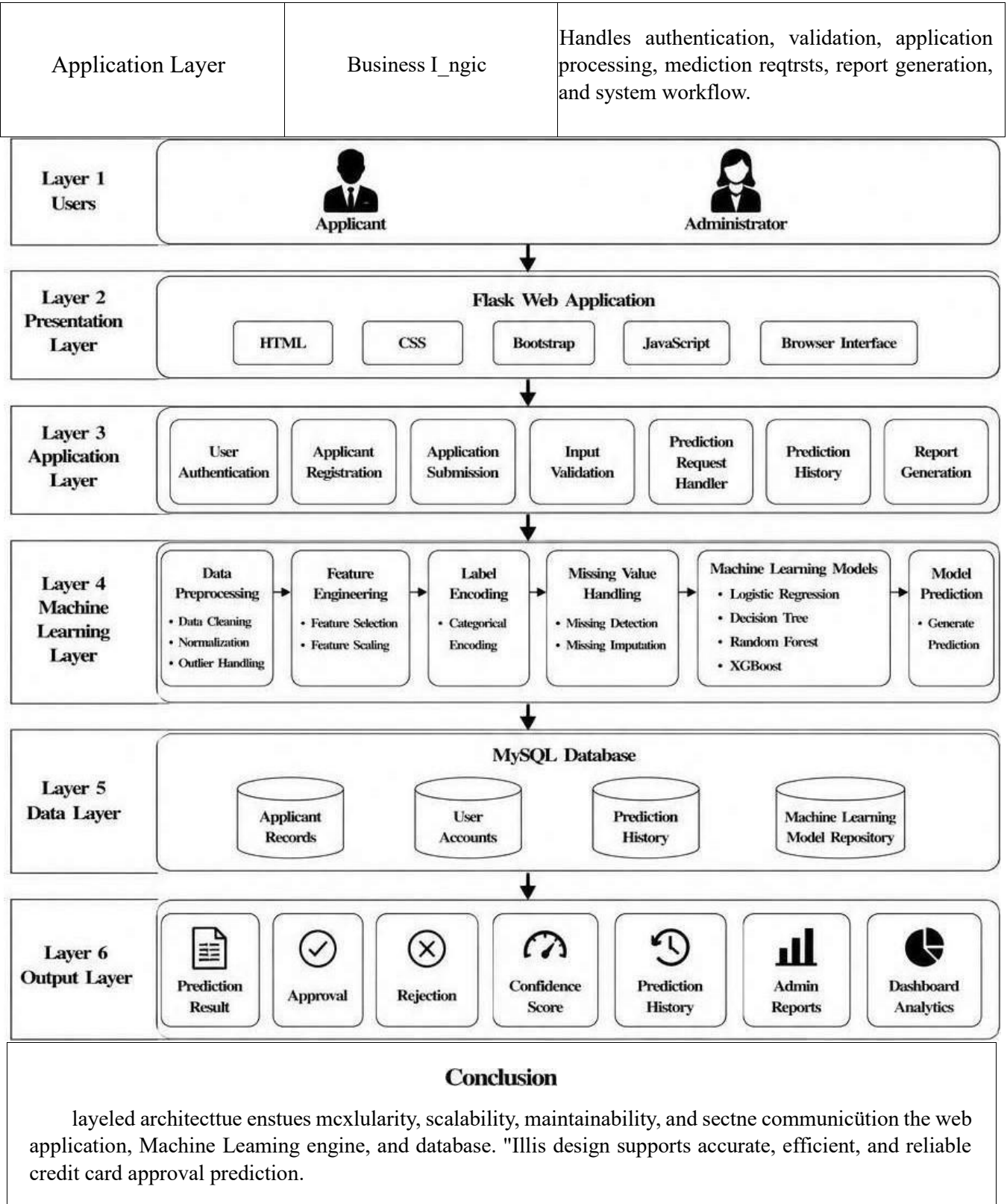
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Project Name	Smart Lender: Machine Learning-Based Loan Approval Prediction System
Maximum Marks	

Solution Architecture

Introduction

The Solution Architecture illustrates the overall structure of the Smart Lender Loan Approval Prediction System. It shows how users interact with the web application, how data flows through preprocessing and Machine Learning components, how loan approval predictions are generated, and how information is securely stored in the database..

Layer	Component	Description
Presentation Layer	Flask Web Interface	Provides responsive web pages for applicant registration, login, application submission, and prediction result display.



Architecture Component Description

The following table describes the Responsibilities of each architectural layer and explains how they interact to provide an intelligent credit card approval prediction system.

Machine Learning Layer	Prediction Engine	Performs preprocessing, feature engineering, model training, classification, and approval prediction using trained Machine Learning algorithms.
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Data Layer	MySQL Database	Stores applicant information, user credentials, prediction history, and application records securely.
Model Storage	Serialized Machine Learning Model	Stores trained Logistic Regression, Decision Random Forest, and XGBoost models for prediction.
Security Layer	Authentication & Authorization	Ensures secure login, encrypted credentials, and role-based access control.
Output Layer	Prediction & Reports	Displays loan approval results, confidence scores, dashboards, and administrative reports
System Advantages ☐ Fast response time ☐ Modular layered architecture ☐ Cloud deployment ready ☐ High scalability ☐ Supports future Machine Learning model updates ☐ Secure user authentication and data protection ☐ Clear separation of responsibilities ☐ Reliable Machine Learning predictions - Easy maintenance and system upgrades		
Conclusion The proposed architecture provides a scalable, modular, and maintainable foundation for the Smart Lender Loan Approval Prediction System. The layered design simplifies future enhancements while ensuring accurate, efficient, and secure loan approval prediction services suitable for real-world financial applications..		